

Matrices

4.5 Linear Dependence and Independence

$$\vec{A} = [A_1 \quad A_2 \quad A_3]$$

$$\vec{B} = [B_1 \quad B_2 \quad B_3]$$

Let vectors be

$$a_{(1)}, a_{(2)}, a_{(3)}, \dots, a_{(m)} \quad (A)$$

Linear combination:- $c_1 a_{(1)} + c_2 a_{(2)} + c_3 a_{(3)} + \dots + c_{(m)} a_{(m)}$

Now Let,

$$c_1 a_{(1)} + c_2 a_{(2)} + c_3 a_{(3)} + \dots + c_{(m)} a_{(m)} = 0 \quad (1)$$

(a) Equation (1) is only true if all c_m 's = 0 then vectors in equation (A) are Linearly Independent.

(b) Ex: $c_1 a_{(1)} + c_2 a_{(2)} = 0 \Rightarrow a_{(2)} = \frac{-c_1 a_{(1)}}{c_2}$, So Linearly Dependent.

$\therefore a_{(2)}$ can be written in terms of $a_{(1)}$, that's why they are linearly dependent.

Linearly independent vectors are truly essential vectors.

Example: Let $a_{(1)} = \{3 \ 0 \ 2 \ 2\}$

$a_{(2)} = \{-6 \ 42 \ 24 \ 54\}$

$a_{(3)} = \{21 \ -21 \ 0 \ -15\}$

$\Rightarrow 6a_{(1)} - \frac{1}{2}a_{(2)} - a_{(3)} = 0$ ($\because c_1, c_2, c_3 \neq 0$) So $a_{(1)}, a_{(2)}$ & $a_{(3)}$ are Linearly dependent.

And $a_{(1)}, a_{(2)}$ are Linearly Independent.